MATH245 (Honours Algebra I) Notes

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Part I

Preliminaries

1 Sets

1.1 Basic Set Theory

Definition 1.1.1

A **set** is a collection of elements.

Intuitively, a set can be thought of as a bag—our set—which may or may not contain stuff—our elements. Sets are the most fundamental objects in mathematics and many important concepts, such as functions and spaces, can be defined using sets.

If we want to say that an element x belongs to some set S, we write

$$x \in S$$

with the "$\in$" symbol meaning "an element of" (so $x \in S$ literally means x is an element of S).

When defining a set, we use curly braces "{}" to enclose the elements inside of it.

Example 1.1.2. The following are some basic examples of sets:

1. $A := \{0, 3, \heartsuit\}$ defines a set, named A, to the elements 0, 3, and $\heartsuit$ (elements do not necessarily need to be numbers, indeed another set can also be an element!)
2. $B := \{\text{all odd integers}\}$ defines a set, named B, to all odd integers (e.g. $-1 \in B$ and $127 \in B$).
3. $C := \{n : n \text{ is an integer satisfying } n^3 = n\}$ (the colon ":" in between the curly braces stands for "such that," so C is the set of all integers n such that $n^3 = n$ (equivalently, $C := \{-1, 0, 1\}$))
4. The standard sets:

   $\mathbb{N} := \{0^1, 1, 2, 3, \dots\}$ (the set of natural numbers)
   $\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$ (the set of integers)
   $\mathbb{Q} := \left\{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\right\}$ (the set of rational numbers)
   $\mathbb{R} :=$ the set of real numbers[2]
   $\mathbb{C} := \left\{a + bi : a, b \in \mathbb{R}, i = \sqrt{-1}\right\}$ (the set of complex numbers)

Remark 1.1.3. Order and repetition in sets do not matter. Indeed, the set $A := \{0, 3, \heartsuit\}$ is equal to the sets $A_1 := \{3, \heartsuit, 0\}$ and $A_2 := \{0, 3, \heartsuit, 0\}$.

[1] Some sources omit 0 from the set of natural numbers, and define $\mathbb{N} := \{1, 2, 3, \dots\}$. However in this class, we will always assume $0 \in \mathbb{N}$, and define $\mathbb{N}^+ := \{1, 2, 3, \dots\}$.

[2] The set of real numbers is surprisingly difficult to define. Take MATH255 to learn how to rigorously define $\mathbb{R}$.

Definition 1.1.4

Two sets A and B are said to be equal if and only if they have the same elements. Symbolically, we write $A = B \iff \forall x, x \in A$ when $x \in B$.

In the above theorem, we introduced two key symbols often used in mathematics. The first, the "$\iff$" symbol, stands for "if and only if." In mathematics, this means that the statement on the left is equivalent to the statement on the right. The second symbol, the "$\forall$", stands for "for all" or "for every" and is called the **universal quantifier**. Symbolic notation is often eschewed in formal writing but is used very often in more casual mathematical writing.

Remark 1.1.5. By Definition 1.0.3, we see that the set A in Remark 1.0.2 is equal to the sets A_1 and A_2.

Definition 1.1.6

Let A and B be arbitrary sets. We say that A **is a subset of** B, or A **is contained in** B, if for every $x \in A$, $x \in B$. In this case we write, $A \subseteq B$, where "$\subseteq$" is the subset symbol[3].

Remark 1.1.7. Note that $A = B$ holds when $A \subseteq B$ and $B \subseteq A$ both hold. This gives us a very useful way to prove if two sets are equal (we simply prove that one set is the subset of the other and vice versa).

Definition 1.1.8

The set with no elements is called the **empty set**, and is denoted $\emptyset := \{\}$.

Intuitively, the empty set can be thought of as a bag with nothing inside of it. The empty set is *not* nothing, but it contains nothing.

Example 1.1.9. We conclude with some examples of sets and some of their properties:

1. For every set A, $\emptyset \subseteq A$.
2. $\emptyset \neq \{\emptyset\}$, indeed we see that $\emptyset \subseteq \{\emptyset\}$ but $\{\emptyset\} \not\subseteq \emptyset$.
3. $\{\emptyset, \{\emptyset\}\} \neq \{\{\emptyset\}\}$
4. In set theory, we write the numbers $0 := \emptyset$, $1 := \{0\}$, $2 := \{0, 1\}$, and so on.

1.2 Basic Set Operations

Now that we have a basic understanding of sets, we will now learn how to work with multiple sets. What do we do if we want to compare elements across different sets? For instance, how can we "add" and "subtract" elements of different sets, or find elements that they have in common? This subsection aims to introduce the fundamental operations on sets that help us answer these questions.

[3]Many textbooks use the symbol "$\subset$" to denote a subset instead of $\subseteq$, i.e. if A is a subset of B, then $A \subset B$ is written. In these notes, I will use the symbol $\subset$ only when one set is a *proper subset* of the other, in the same way that $<$ defines a strict inequality.

Definition 1.2.10

Let A and B be two sets. Define:

- Their **union**, denoted $A \cup B$, as the set $\{x : x \in A \text{ or } x \in B\}$
- Their **intersection**, denoted $A \cap B$, as the set $\{x : x \in A \text{ and } x \in B\}$
- Their **difference**, denoted $A \setminus B$, as the set $\{x : x \in A \text{ and } x \notin B\}$.

If A is an ambient set (i.e. there are other sets that are contained in A), then if $B \subseteq A$ we say the **complement** of B relative to A, denoted B^c, is the difference:

$$B^c := A \setminus B$$

Example 1.2.11. Within $\mathbb{Z}$, define $A := \{n \in \mathbb{Z} : n \text{ is even}\}$. We have that the complement of A relative to $\mathbb{Z}$, $A^c := \{n \in \mathbb{Z} : n \text{ is odd}\}$.

Definition 1.2.12

For two elements a and b (a and b are not necessarily sets), we define their **ordered pair**, denoted (a, b), to be the tuple satisfying the property:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

In particular, if $a \neq b$, then $(a, b) \neq (b, a)$.

We will use this notion of ordered pairs to help us with the following important definition:

Definition 1.2.13

Let A, B be two sets. We define their **Cartesian product**, denoted $A \times B$, as the set of ordered pairs $\{(a, b) : a \in A \text{ and } b \in B\}$.

A notable Cartesian product is the Euclidean plane $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$. You have already seen this plane defined as the xy-plane, where each ordered pair (x, y) represents the x- and y-coordinates of a point on the plane.

Example 1.2.14. Let $A := \{0, 1, 2\}$ and $B := \{a, b\}$. Then their Cartesian product $A \times B$ is the set $\{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b)\}$.

Remark 1.2.15. In general, if A has n elements and B has m elements, then their cartesian product $A \times B$ has $n \cdot m$ elements.

Definition 1.2.16

Let A be a set. We say the **power set of** A, denoted 2^A or $\mathcal{P}(A)$, is the set of all subsets of A, which includes A and the empty set $\emptyset$.

Example 1.2.17. Let $A := \{0, 1, 2\}$, then its power set $\mathcal{P}(A)$ is:

$$\mathcal{P}(A) := \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$$

Remark 1.2.18. For each element in a nonempty set, we can either choose to include or not include it in a subset of A. This means that for a set A with n elements, its power set has 2^n elements.

1.3 Algebra of Set Operations

We will show our first proposition (a statement which we will prove) here.

Proposition 1.3.19

Let A, B, and C be arbitrary sets. Then the following properties hold:

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
3. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
4. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$
5. Given an ambient set U such that $A \subseteq U$, $(A^c)^c = A$

Proof. Each statement in this proposition asserts that two sets are equal. By Remark 1.0.6, we prove each equality by showing that each side is a subset of the other.

Let us start by proving (1). We first show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$. Let $x \in A \cap (B \cup C)$. By definition, $x \in A$ and $x \in B \cup C$. Therefore, $x \in B$ or $x \in C$.

If $x \in B$, then $x \in A \cap B$, and hence $x \in (A \cap B) \cup (A \cap C)$. If $x \in C$, then $x \in A \cap C$, and hence $x \in (A \cap B) \cup (A \cap C)$. Therefore,

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C).$$

Conversely, let $x \in (A \cap B) \cup (A \cap C)$. Then $x \in A \cap B$ or $x \in A \cap C$.

If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $x \in A \cap (B \cup C)$. If $x \in A \cap C$, then $x \in A$ and $x \in C$, so $x \in A \cap (B \cup C)$. Therefore,

$$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C).$$

Hence condition (1) holds.

For (2), let $x \in A \cup (B \cap C)$. If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$. Therefore, $x \in (A \cup B) \cap (A \cup C)$.

If $x \in B \cap C$, then $x \in B$ and $x \in C$. Thus $x \in A \cup B$ and $x \in A \cup C$, so again $x \in (A \cup B) \cap (A \cup C)$. Hence

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C).$$

Conversely, let $x \in (A \cup B) \cap (A \cup C)$. Then $x \in A \cup B$ and $x \in A \cup C$.

If $x \in A$, then $x \in A \cup (B \cap C)$. Otherwise, $x \notin A$. Since $x \in A \cup B$ and $x \notin A$, we must have $x \in B$. Similarly, since $x \in A \cup C$ and $x \notin A$, we must have $x \in C$. Thus $x \in B \cap C$, and therefore $x \in A \cup (B \cap C)$. Hence

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C),$$

so condition (2) holds.

For (3), let $x \in A \setminus (B \cup C)$. By definition, $x \in A$ and $x \notin B \cup C$. Since $x \notin B \cup C$, we have $x \notin B$ and $x \notin C$; otherwise, x would belong to $B \cup C$. Thus $x \in A \setminus B$ and $x \in A \setminus C$. Therefore,

$$x \in (A \setminus B) \cap (A \setminus C),$$

and hence

$$A \setminus (B \cup C) \subseteq (A \setminus B) \cap (A \setminus C).$$

Conversely, let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A \setminus B$ and $x \in A \setminus C$, which means $x \in A$, $x \notin B$, and $x \notin C$. Since $x \notin B$ and $x \notin C$, it follows that $x \notin B \cup C$. Thus $x \in A \setminus (B \cup C)$. Therefore,

$$(A \setminus B) \cap (A \setminus C) \subseteq A \setminus (B \cup C),$$

and condition (3) holds.

For (4), let $x \in A \setminus (B \cap C)$. By definition, $x \in A$ and $x \notin B \cap C$. Therefore, $x \notin B$ or $x \notin C$.

If $x \notin B$, then $x \in A \setminus B$. If $x \notin C$, then $x \in A \setminus C$. Hence $x \in (A \setminus B) \cup (A \setminus C)$. Therefore,

$$A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C).$$

Conversely, let $x \in (A \setminus B) \cup (A \setminus C)$. Then $x \in A \setminus B$ or $x \in A \setminus C$. Thus $x \in A$ and either $x \notin B$ or $x \notin C$. Therefore, $x \notin B \cap C$, so $x \in A \setminus (B \cap C)$. Hence

$$(A \setminus B) \cup (A \setminus C) \subseteq A \setminus (B \cap C),$$

and condition (4) holds.

Finally, by definition of the complement relative to U,

$$A^c = U \setminus A.$$

Therefore,

$$(A^c)^c = U \setminus (U \setminus A) = A.$$

Hence condition (5) holds. ■

Proposition 1.3.20 (De Morgan's Laws)

Let U be a set, and let $B, C \subseteq U$. Then:

1. $(B \cap C)^c = B^c \cup C^c$.
2. $(B \cup C)^c = B^c \cap C^c$.

Proof. For (1), let $x \in (B \cap C)^c$. By definition of the complement, $x \in U$ and $x \notin B \cap C$. Therefore, it is not the case that both $x \in B$ and $x \in C$. Hence $x \notin B$ or $x \notin C$.

If $x \notin B$, then $x \in B^c$. If $x \notin C$, then $x \in C^c$. In either case, $x \in B^c \cup C^c$. Thus

$$(B \cap C)^c \subseteq B^c \cup C^c.$$

Conversely, let $x \in B^c \cup C^c$. Then $x \in B^c$ or $x \in C^c$, so $x \notin B$ or $x \notin C$. Therefore, $x \notin B \cap C$. Since $x \in U$, it follows that $x \in (B \cap C)^c$. Hence

$$B^c \cup C^c \subseteq (B \cap C)^c.$$

Therefore,

$$(B \cap C)^c = B^c \cup C^c.$$

For (2), let $x \in (B \cup C)^c$. By definition, $x \in U$ and $x \notin B \cup C$. Therefore, $x \notin B$ and $x \notin C$. Thus $x \in B^c$ and $x \in C^c$, which implies that

$$x \in B^c \cap C^c.$$

Hence

$$(B \cup C)^c \subseteq B^c \cap C^c.$$

Conversely, let $x \in B^c \cap C^c$. Then $x \in B^c$ and $x \in C^c$, meaning that $x \notin B$ and $x \notin C$. Therefore, $x \notin B \cup C$. Since $x \in U$, we conclude that $x \in (B \cup C)^c$. Thus

$$B^c \cap C^c \subseteq (B \cup C)^c.$$

Therefore,

$$(B \cup C)^c = B^c \cap C^c.$$ ■

2 Methods of Proof

Propositions 1.3.19 and 1.3.20 were two statements with results that we did not know were true, but were able to deduce as true by *proving* them. We proved the statements were true by showing that each side of the equality was a subset of the other. This method of proof outlined in Remark 1.1.7 works well when trying to prove equality, but there are many, many other methods of proof that we can use to show other statements are true.

2.1 Proof by Contradiction

Suppose we are asked to prove a statement P. We can prove P is true by assuming that P is false (i.e $\neg P$—the "$\neg$" symbol means "not"—is true) and then showing that $\neg P$ cannot be true. This is called a **proof by contradiction.**

The following is one of the most fundamental statements in mathematics and elengantly proved by contradiction.

Proposition 2.1.1

$\sqrt{2} \notin \mathbb{Q}$ ($\sqrt{2}$ is not a rational number).

Proof. Let us assume, by contradiction, that the negation of the proposition is true, i.e. we assume that $\sqrt{2} \in \mathbb{Q}$. Then, we can write $\sqrt{2} = p/q$, where p and q are two integers, $q \neq 0$, that are irreducible (i.e. they have no common factors). Then $2 = p^2/q^2$ and $2q^2 = p^2$. Recall an even integer a can be writen as $a := 2b$ for $b \in \mathbb{Z}$. Since p^2 is the product of 2 and q^2, then p^2 is even, which immediately implies p is even. Since p is even, it can be written as $p = 2k$ where $k \in \mathbb{Z}$. Squaring both sides yields $p^2 = 4k^2$. Let $2q^2 = 4k^2$ and so $q^2 = 2k^2$, so q^2 is even which implies q^2 is also even, but we assumed that p and q are irreducible. Contradiction. So $\sqrt{2} \notin \mathbb{Q}$. ■

2.2 Proof by Contrapositive

Suppose we are trying to prove a statement $P \implies Q$, i.e. if P is true, then Q is true as well. This statement is logically equivalent to

$$\neg Q \implies \neg P$$

which is the **contrapositive** of $P \implies Q$. Note that this differs from a proof by contradiction, as instead of leading to an absurdity, proving by the contrapositive shows $\not P$ is true and thus the statement must be true. It is often easier to prove the contrapositive of the statement rather than proving the statement directly.

Lemma 2.2.2

Let $a, b \in \mathbb{Z}$. If ab is even, then a or b are even.

Proof. Let us prove the contrapositive of the statement: if both a and b are odd, then ab is odd. Assume a and b are odd integers, then $a := 2k + 1$ and $b := 2l + 1$ for $k, l \in \mathbb{Z}$. We write $ab = (2k+1)(2l+1) = 4kl+2k+2l+1 = 2(2kl+k+l)+1$. Let $m := 2kl+k+l$, we have $ab = 2m+1$ and since $m \in \mathbb{Z}$, ab is odd. ■

2.3 Proof by Induction

A key property of the natural numbers is that they are well ordered, namely every nonempty subset $S \subseteq \mathbb{N}$ has a unique least element n_0 (i.e. every element in S is less than or equal to n_0. This property is known as the **axiom of well-orderedness**. The axiom of well-orderedness gives rise to an equivalent, but extremely useful statement:

Theorem 2.3.3 (Principle of Induction)

Let $P \subseteq \mathbb{N}$. Suppose the following statements are true:

1. For $n_0 \in \mathbb{N}$, $n_0 \in P$ (the base case).
2. For all $n \in \mathbb{N}$, if $n \in P$, then $n + 1 \in P$ (the inductive step).

Then $P \supseteq \{n \in \mathbb{N} : n \geq n_0\}$.

Proof. By contradiction, suppose that $P \not\supseteq \{n \in \mathbb{N} : n \geq n_0\}$. Then the set $S := \{n \in \mathbb{N} : n \geq 0 \text{ and } n \notin P\} \neq \emptyset$. Since $S \subseteq \mathbb{N}$, there exists a least element $s_0 \in S$ by the well-ordered axiom. We have that $s_0 \neq n_0$ since $n_0 \in P$. Thus $s_0 > n_0$ and $s_0 - 1 \geq n_0$. Since s_0 is a least element of S, $s_0 - 1 \notin S$ hence $s_0 - 1 \in P$. But by the inductive step, $s_0 \in P$. Contradiction. ■

Example 2.3.4. We claim, for any $n \in \mathbb{N}$:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Proof. We will proceed by induction. We first consider the base case $n = 0$:

$$0 = \frac{0(1)}{2} = 0$$

Now, let us assume that the condition holds for every $n \in \mathbb{N}$. We want to show the condition holds for $n + 1$. We now want to prove:

$$1 + 2 + \cdots + n + (n+1) = \frac{(n+1)(n+2)}{2}$$

By the inductive hypothesis, let us write $1 + 2 + \cdots + n = n(n+1)/2$. We can do this because we assumed that the statement holds for every $n \in \mathbb{N}$. With simple algebraic manipulations, we see that:

$$\begin{aligned} \frac{n(n+1)}{2} + (n+1) &= \frac{(n+1)(n+2)}{2} \\ \frac{n(n+1) + 2(n+1)}{2} &= \frac{(n+1)(n+2)}{2} \\ \frac{(n+1)(n+2)}{2} &= \frac{(n+1)(n+2)}{2} \end{aligned}$$

Hence the statement holds by induction. ■

It is worth noting that the well-ordered axiom and Theorem 2.3.3 are equivalent to each other. Indeed, we can prove the well-ordered axiom using the Principle of Induction:

Theorem 2.3.5 (Well-ordered Axiom)

Assume Theorem 2.3.3 is true. Then every nonempty set $S \subseteq \mathbb{N}$ contains a least element.

Proof. We will prove the contrapositive of this statement: if a set $S \subseteq \mathbb{N}$ does not have a least element, then S is an empty set. Let $S \subset \mathbb{N}$ not have a least element. We claim by induction, for all $n \in \mathbb{N}$, $n \notin S$, i.e. $S = \emptyset$.

First consider the base case $n = 0$. Obviously $0 \notin S$ since 0 is the least element of $\mathbb{N}$. Now assume that for every $n \in \mathbb{N}$, $n \notin S$. We have that $n + 1 \notin S$ since it would be the least element in S by the induction hypothesis. So $S = \emptyset$. ■

2.4 Complete Induction

Complete induction is a variant of the Principle of Inuction. To differentiate it from Theorem 2.3.3, complete induction is sometimes called *strong induction* while Theorem 2.3.3 is called *weak induction*. Note that complete induction is logically equivalent to weak induction and by consequence, the well-ordered axiom.

Theorem 2.4.6 (Complete Induction)

Let $n_0 \in \mathbb{N}$ and $P \subseteq \mathbb{N}$ satisfying that for each $n \geq n_0$, if $\{n_0, n_0 + 1, \ldots, n - 1\} \subseteq P$, then $n \in P$. Then $P \supseteq \{n_0, n_0 + 1, \ldots\} = \{n \in \mathbb{N} : n \geq n_0\}$.

Proof. We will prove Theorem 2.4.6 using the Well-ordered axiom. First, remark that for $n = n_0$, $\{n_0, n_0 + 1, \ldots, n - 1\} = \emptyset$. Suppose by contradiction that $P \not\supseteq \{n \in \mathbb{N} : n \geq n_0\}$. Then the set $S := \{n \in \mathbb{N} : n \geq n_0, n \notin P\} \neq \emptyset$ and contains a least element n_α by the axiom of well-orderedness. Remark $n_\alpha > n_0$, since if $n_a = n_0$, we have that $\{n_0, n_0 + 1, \ldots, n - 1\} = \emptyset \implies n \in P$, which would be a contradiction. Hence $n_\alpha - 1 \geq n_0$ and moreover $n_\alpha - 1 \notin S$ since n_α was the least element of S, whence $n_\alpha - 1 \in P$. Indeed, we have that $\{n_0, n_0 + 1, \ldots n_\alpha - 1\} \subseteq P$ by the minimality of n_α, but this implies that $n_\alpha \in P$, which is a contradiction. ∎

Although Theorems 2.3.3 and 2.4.6 are equivalent, there are times (see Proposition 2.5.8 below) where it is much easier to prove a statement by strong induction than by weak induction.

2.5 The Pigeonhole Principle

The Pigeonhole Principle is a powerful consequence of the well-ordered axiom which states that if there are n pigeons and m pigeonholes, where $n \geq m$, then at least one pigeonhole has two or more pigeons in it.

Proposition 2.5.7

Let $a_1, a_2, \ldots a_5 \in \mathbb{Z}$. If we divide all integers by 4, then there are two among them with the same remainder.

Proof. The possible remainders are $\{0, 1, 2, 3\}$. Since there are 5 integers that can have 4 different remainders, we invoke the pigeonhole principle to show that at least two among $a_1, \ldots a_5$ have the same remainder. ∎

Proposition 2.5.8

For any group of of 2 or more people, there are two different people who have the same number of friends.

Proof. Let us first consider the case $n = 2$. The statement is trivial, as both people are either friends with each other or have no friends.

Now let $N \geq 2$ and suppose the statement holds for all $n \leq N$. We consider a group of $N + 1$ friends. Suppose that one person in the group does not have any friends. Then there are N people remaining who have at least one friend. By complete induction, there are at least two people in this group with the same number of friends, so the statement holds for the group of $N + 1$ people.

Suppose now that everyone in the group has at least one friend. Then each person has anywhere from 1 to N friends. Since there are $N + 1$ people in this group, the pigeonhole principle states that there must be two people with the same number of friends. ∎

3 Functions

3.1 Basic Functions

Intuitively, we can think of functions as mapping elements from one set to a unique element in another. More formally:

Definition 3.1.1

A **function** from a set X to a set Y, denoted $f: X \to Y$, is a subset $f \subseteq X \times Y$ satisfying that for each $x \in X$, there exists a unique $y \in Y$ such that the ordered pair $(x, y) \in f$. If this is the case we write $f(x) = y$. The set X is called the **domain** of f while the set Y is called the **codomain** of f.

Example 3.1.2. The simplest function is called the **identity function**, which simply sends each element of a set X to itself. It can be written as $1_A: A \to A$, where $1_A(x) = x$ for all $x \in A$.

Definition 3.1.3

Let $f: X \to Y$ be a function. The **graph** of a f, often denoted Γ_f or simply Γ, is defined by set $\{(x, f(x)) \in X \times Y : f(x) = y\}$.

Example 3.1.4. Consider the function $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) := x^2$. Consider as well the function $g: \mathbb{R} \to [0, \infty)$ given by $g(x) := x^2$. We have that $\Gamma_f = \Gamma_g$, but f and g are technically two different functions since their codomains differ. However, $f \equiv g$.

Definition 3.1.5

For a function $f: X \to Y$ and $A \subseteq X$, $B \subseteq Y$:

1. For $a \in A$, we define the **image of** A **under** f, denoted $f(A) := \{f(a) : a \in A\}$.
2. For $x \in X$, we define the **preimage of** B, denoted $f^{-1}(B) := \{x \in X : f(x) \in B\}$.

Remark 3.1.6. For $f: X \to Y$, $A \subseteq X$ and $B \subseteq Y$, $f^{-1}(f(A)) \supseteq A$, but $f^{-1}(f(A)) \neq A$. Indeed, if we consider the function $f: \mathbb{Z} \to \mathbb{Z}$ given by $f(n) := n^2$, then $f(\mathbb{N}) = \{0, 1, 4, 9, \dots\}$ but $f^{-1}(f(\mathbb{N})) = \mathbb{Z}$. Similarily, $B \supseteq f(f^{-1}(B))$ but $f(f^{-1}(B)) \neq B$. Indeed, for f given as above, $f^1(\mathbb{Z}) = \mathbb{Z}$, but $f(f^{-1}(\mathbb{Z})) = f(\mathbb{Z}) = \{0, 1, 4, 9, \dots\}$.

Definition 3.1.7

Let $f: X \to Y$ be a function.

1. We say f is **injective** or **one-to-one** if for every $y \in Y$, there exists one and only one $x \in X$ such that $f(x) = y$. In other words, for all $x, x_0 \in X$, $f(x) = f(x_0) \implies x = x_0$.
2. We say f is **surjective** or **onto** if for all $y \in Y$, there exists $x \in X$ such that $f(x) = y$. In other words $f(X) = Y$.
3. We say f is **bijective** if it is both injective and surjective.

Example 3.1.8. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) := x^2$ is neither injective nor surjective, The function $g: \mathbb{R} \to [0, \infty)$, $g(x) := x^2$ is surjective but not injective, as $f(1) = f(-1)$ but $1 \neq -1$.

The function $h\colon [0,\infty) \to [0,\infty)$ defined by $h(x) := x^2$ is both injective and surjective, hence it is bijective.

3.2 Compositions of Functions

Definition 3.2.9

Let $f\colon X \to Y$ and $g\colon Y \to Z$ be two functions. The **composition of f and g**, denoted $g \circ f\colon X \to Z$, is the function denoted by $g \circ f := g(f(x))$.

Remark 3.2.10. A composition of functions is associative, that is, if $f\colon X \to Y$, $g\colon Y \to Z$, $h\colon Z \to W$, then $h \circ (g \circ f) = (h \circ g) \circ f$.

Example 3.2.11. For $f\colon \mathbb{N} \to \mathbb{N}$, $n \mapsto n+1$ and $g\colon \mathbb{N} \to \mathbb{N}$, $n \mapsto n^2$, then $g \circ f := g(f(n)) = g(n+1) = (n+1)^2$, and $f \circ g := f(g(n)) = f(n^2) = n^2 + 1$.

Proposition 3.2.12

Let $f\colon X \to Y$, $g\colon Y \to Z$ be two functions and $g \circ f\colon X \to Z$ be their composition.

1. If $g \circ f$ is injective, then f is injective.
2. If $g \circ f$ is surjective, then g is surjective.

Proof. First assume $g \circ f$ is injective. Let $x, x' \in X$ such that $f(x) = f(x')$. Since $g \circ f$ is injective:

$$(g \circ f)(x) = (g \circ f)(x') \iff g(f(x)) = g(f(x')) \implies x = x'$$

Now assume $g \circ f$ is surjective. Let $z \in Z$ and we want to show there exists a $y \in Y$ such that $g(y) = z$. Since $g \circ f$ is surjective, there exists an $x \in X$ such that $(g \circ f)(x) = z$, i.e. $g(f(x)) = z$. Let $y = f(x)$ and we are done. ∎

Definition 3.2.13

Let $f\colon X \to Y$. Let $g\colon Y \to X$.

1. We say g is the **right inverse of** f if $f \circ g = 1_X$.
2. We say g is the **left inverse of** f if $g \circ f = 1_X$.
3. We say g is the **inverse of** f (sometimes called a **two sided inverse of** f) if it is both a left and right inverse of f.

Proposition 3.2.14

Let $f\colon X \to Y$ be a function. If $g_l\colon Y \to X$ is a left inverse and $g_r\colon Y \to X$ is a right inverse, then $g_l \equiv g_r$ and thus is a two-sided inverse.

Proof. We can write:

$$g_l = g_l \circ 1_Y = g \circ (f \circ g_r) = (g_l \circ f) \circ g_r = 1_X \circ g_r = g_r$$ ∎

Proposition 3.2.15

Let $f : X \to Y$ be a function where $X \neq \emptyset$.
1. f admits an inverse if and only if f is bijective.
2. f admits a left inverse if and only if f is injective.
3. f admits a right inverse if and only if f is surjective.

Proof. Left as an exercise in the assignment. ■

Remark 3.2.16. If $f : X \to Y$ has an inverse, then it is unique and we denote it by $f^{-1} : Y \to X$, such that $f^{-1}(y) = x$ if $f(x) = y$.

3.3 Binary Relations and Equivalence Relations

Definition 3.3.17

A **binary relation** on a set X is a an arbitrary subset of $X \times X$.

We are interested in a more general class of binary relations:

Definition 3.3.18

A **partial order** on a set, denoted $\leq$ is a binary relation where:
1. $\leq$ is reflexive: $(x, x) \in \leq$, i.e. $x \leq x$ for all $x \in X$.
2. $\leq$ is assymmetric: for all $x, y \in X$ such that $x \neq y$,$x \leq y$ but $y \not\leq x$.
3. $\leq$ is transitive: for all $x, y, z \in X$, if $x \leq y$ and $y \leq z$, then $x \leq z$.

Remark 3.3.19. On $\mathbb{R}$, the usual "$\leq$" is a partial order; in fact it is a linear, or total order, i.e. for all $x, y \in X$, $x \leq y$ and $y \leq x$.

However, the most interesting—and important—class of binary relations are:

Definition 3.3.20

An **equivalence relation** on a set X is a binary relation E satisfying:
1. Reflexivity: For all $x \in X$, xEx (any element in X is equivalent to itself).
2. Symmetry: For all $x, y \in X$, if xEy, then yEx.
3. Transitivity: For all $x, y, z \in X$, if xEy and yEz, then xEz.

Example 3.3.21. Let $\sim$ be an equivalence relation on $\mathbb{Z}$, where $x \sim y$ when $x - y$ is even. This is a valid equivalence relation. Indeed, $x \sim x$ since $x - x = 0$ which is even, so $\sim$ is symmetric. Moreover, if $x \sim y$, i.e. $x - y = b$ for some even $b \in \mathbb{Z}$, then $y - x = -(x - y) = -b$ which is still even, so $y \sim x$ and $\sim$ is symmetric. Lastly, if $x \sim y$ and $y \sim z$, i.e. $x - y = b$ and $y - z = c$ where b and c are even integers, then $x - z = (x - y) + (y - z) = b + c$ which is even. So $x \sim z$ thus $\sim$ is transitive.

Definition 3.3.22

Let E be an equivalence relation on a set X. For all $x \in X$, the E-**class**, or **equivalence class** of x is the set

$$[x]_E := \{y \in X : xEy\}.$$

Proposition 3.3.23

Let $\sim$ be an equivalence relation on a set X. For all $x, y \in X$, the following are true:
(a) $x \sim y \iff [x]_\sim = [y]_\sim$.
(b) $[x]_\sim$ and $[y]_\sim$ are either disjoint or equal, i.e. either $[x]_\sim = [y]_\sim$ or $[x]_\sim \cap [y]_\sim = \emptyset$.

Proof. For (a), let us first assume $x \sim y$. Let $z \in [y]_\sim$, so $y \sim z$, so my transitivity $x \sim z$, so $z \in [x]_\sim$, i.e. $[y]_\sim \subseteq [x]_\sim$. By symmetry, we also have $y \sim x$, so analogously we get $[x]_\sim \subseteq [y]_\sim$. Conversely, suppose $[x]_\sim = [y]_\sim$. Then by reflexivity, $y \in [y]_\sim$, thus $y \in [x]_\sim$, hence $x \sim y$.

For (b), suppose $[x]_\sim$ and $[y]_\sim$ are not disjoint, i.e. there exists $z \in [x]_\sim \cap [y]_\sim$ and show that $[x]_\sim = [y]_\sim$. By the definition of z, so $x \sim z$ and $y \sim z$. By (a), $[x]_\sim = [z]_\sim$ and $[y]_\sim = [z]_\sim$, hence $[x]_\sim = [y]_\sim$. ∎

Definition 3.3.24

Let $\sim$ be an equivalence relation on a set X. The **quotient of** X **by** $\sim$, denoted $X/\sim$, is the set of all equivalence classes:

$$X/\sim := \{[x]_\sim : x \in X\}.$$

4 Number Systems

4.1 Constructing The Rationals

We suppose that $\mathbb{N}$ is given by the set theory axioms. Then we can define $\mathbb{Z}$ as the union $\mathbb{N} \cup \{-n : n \in \mathbb{N}^+\}$ where $\mathbb{N}^+$ denotes the positive natural numbers. We can then define the order and multiplication and addition on $\mathbb{Z}$ the usual way though the equivalence classes $\leq$, $+$ and $\cdot$ on $\mathbb{N}$.

We can now define the set of rationals $\mathbb{Q}$. Begin by declaring $Q_0 := \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$. $Q_0 \neq \mathbb{Q}$, indeed $(1, 2) \neq (3, 6)$ while we know $\frac{1}{2} = \frac{3}{6}$. To fix this, we define an equivalence relation on Q_0 by:

$$(a, b) \ (c, d) \iff ad = bc$$

One can easily check that is an equivalence relation, and we denote the -class of (a, b) by:

$$\frac{a}{b} := [(a, b)]$$

Now by definition, $\frac{1}{2} = \frac{3}{6}$. Denote the quotient by $\mathbb{Q}$, i.e. $\mathbb{Q} := Q_0/$. Define $+$ on $\mathbb{Q}$ as a function $+ : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ by:

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}$$

We now show that + is well-defined, i.e. does not depend on the choices ofrepresentatives: let $\frac{a'}{b'} = \frac{a}{b}$ and $\frac{c'}{d'} = \frac{c}{d}$ and we show that:

$$\frac{a'd' + b'c'}{b'd'} = \frac{ad + bc}{bd}$$

The verification of this is an elementary exercise.

Define $\cdot$ and $\leq$ in a similar manner.

Remark 4.1.1. Once this definition is understood, elements of $\mathbb{Q}$ may be thought of as special points on the real line as opposed to equivalence classes.

4.2 An Overview of Real Numbers

As we know from elementary math, each rational number admits a decimal representaton. For instance $\frac{1}{3} = 0.3333\ldots$, which is eventually periodic. We can define $\mathbb{R}$ as the set of all such decimal representations, i.e. as a set of sequences of digits $\{0, 1, \ldots, 9\}$ with a period as one of the entries, and the first entry being the sign. However, this is not quite correct, say, $01.2000\ldots$ should be the same as $1.2000\ldots$, also $0.999\ldots$ should be the same as $1.000\ldots$, so we need to identify them using an equivalence relation. One can define $\leq$, + and $\cdot$ in expected ways using addition and multiplication algorithms learnt in school.

4.3 The Field of Complex Numbers

Definition 4.3.2

We define the set of **complex numbers**, denoted $\mathbb{C}$, as the set $\{a + ib : a, b \in \mathbb{R}\}$ where $i = \sqrt{-1}$ and is known as the **imaginary solution** to $x^2 = -1$. Elements of $\mathbb{C}$ are called **complex numbers**.

We define + and $\cdot$ on $\mathbb{C}$ using that $i^2 = -1$ and the usual distributivity laws:

1. $(a + ib) + (c + id) := (a + c) + i(b + d)$
2. $(a + ib) \cdot (c + id) := (ac - bd) + i(ad + bc)$

There is no natural order on $\mathbb{C}$ which respects + and $\cdot$, so we do not sefine $\leq$ on $\mathbb{C}$.

Furthermore, for $z \in \mathbb{C}$ where $z := a + ib$, define the **real part of** z, denoted $\mathrm{Re}(z) := a$ and the **imaginary part of** z, denoted $\mathrm{Im}(z) := b$. We also write the absolute value of z as:

$$|z| := \sqrt{|\mathrm{Re}(z)|^2 + |\mathrm{Im}(z)|^2}$$

which is markedly similar to the notion of absolute values in Euclidean space. Hence we can interpret $\mathbb{C}$ geometrically as being akin to $\mathbb{R}^2$ where the x-coordinate is $\mathrm{Re}(z)$ and the y-coordinate is the imaginary part $\mathrm{Im}(z)$, and the distance between points is the Euclidean distance:

$$|z - z'| := \sqrt{|\mathrm{Re}(z) - \mathrm{Re}(z')|^2 + |\mathrm{Im}(z) - \mathrm{Im}(z')|^2}$$

Definition 4.3.3

Let $z \in \mathbb{C}$. We say the **conjugate of** z, denoted $\overline{z} := \mathrm{Re}(z) - \mathrm{Im}(z)$ so for $z := a + ib$, $\overline{z} = a - ib$.

Observe that $z \cdot \overline{z} = |z|^2$.

Definition 4.3.4

Let $z \in \mathbb{C} \setminus \{0\}$. Define the **sign of** z, denoted $\mathrm{sgn}(x)$, as:

$$\mathrm{sgn}(z) := \frac{z}{|z|}$$

so $|\mathrm{sgn}(z)| = 1$.

Observe, since $|\mathrm{sgn}(z) = 1|$, we have $\mathrm{sgn}(z) = \cos\theta + i\sin\theta =: e^{i\theta}$ for some $\theta \in \mathbb{R}$. This identity is called *Euler's equation*, and we define $\arg(z) := \{\theta \in \mathbb{R} : \mathrm{sgn}(z) = e^{i\theta}\}$. Additionally, we can write $\arg(z) = \{\theta_n + 2\pi k : k \in \mathbb{Z}\}$ for some $\theta_n \in [0, 2\pi)$.

Thus it follows that we can write any $z \in \mathbb{C}$ in polar form as:

$$z = re^{\{i(\theta) + 2\pi k)\}}$$

for all $k \in \mathbb{Z}$, where $r := |z|$ and $\theta \in [0, 2\pi)$.

Since $i = \sqrt{-1}$ and is a root to the equation $x^2 + 1 = 0$, we can define complex roots of polynomials that do not have real roots. This gives rise to the following theorem which we will revisit later:

Theorem 4.3.5 (Fundamental Theorem of Algebra)

Let $p(x) := a_0 + a_1 x + \cdots + a_n x^n$ be a nonconstant polynomial over $\mathbb{C}$, i.e. $a_0, \ldots, a_n \in \mathbb{C}$, $n \geq 1$ and $a_n \neq 0$. Then $p(x)$ has at least one root in $\mathbb{C}$.

Part II

Arithmetic of Integers

5 Division and its Consequences

5.1 Divisibility

Definition 5.1.1

Let $a, b \in \mathbb{Z}$. We say that a **divides** b, denoted $a \mid b$, if there exists $q \in \mathbb{Z}$ such that $b = aq$.

Remark 5.1.2. Definition 5.1.1 differs slightly from our usual assumptions of divisibility. For instance—by this definition—zero divides zero, i.e. $0 \mid 0$, since we can choose $q = 0$ to satisfy $0 = 0$. Obviously we know that $0/0$ is indeterminate.

Proposition 5.1.3

Let $a, b, c \in \mathbb{Z}$. The following statements hold:
(a) $a \mid b \implies a \mid -b$.
(b) $a \mid b \implies a \mid bc$.
(c) $a \mid b$ and $a \mid c \implies a \mid (b \pm c)$.
(d) $a \mid b$ and $b \mid c \implies a \mid c$.

Proof. The proof of (a) is trivial (use $-q$). For (b), if $a \mid b$, then there exists a $q \in \mathbb{Z}$ such that $b = qa$. Multiply both sides by c and let $r = qc \in \mathbb{Z}$, then $a \mid bc$. For (c), we have that there exists q and $r \in \mathbb{Z}$ such that $b = qa$ and $c = ra$. So $b \pm c = qa \pm ra = a(r \pm q)$. Let $s := r \pm q \in \mathbb{Z}$ and so $a \mid (b \pm c)$. For (d), by assumption there exists q and $r \in \mathbb{Z}$ such that $b = qa$ and $c = rb$, so $c = rqa$. Let $s := rq \in \mathbb{Z}$, so $a \mid c$. ∎

5.2 Remainders

However, as we know very well, most numbers do not cleanly divide other numbers. What remains after division is called the *residue*, or more commonly, the *remainder*. The following theorem formalizes this idea.

Theorem 5.2.4 (Quotient and Remainder Theorem)

Let $a, b \in \mathbb{Z}$ where $b \neq 0$. Then there exists unique integers $q, r \in \mathbb{Z}$ such that $a = bq + r$ and $0 \leq r < |b|$.

Proof. We must show existence and uniqueness of q and r in this proof. We begin by showing that q and r exist. Define:

$$\begin{aligned} S &:= \{r \in \mathbb{N} \colon r = a - bq, q \in \mathbb{Z}\} \\ &= \{a - bq \colon q \in \mathbb{Z}, a - bq \geq 0\} \end{aligned}$$

Then $S \subseteq \mathbb{N}$ and $S \neq \emptyset$. Indeed, if $a \geq 0$, then taking $q := 0$ gives $a = a - bq \in S$, and if $a < 0$ then take $q := \pm a$ with sign chosen such that $-bq \geq 0$, then $a - bq = a + |b||a| \geq a + |a| \geq 0$, whence $a + bq \in S$. By the well-ordering axiom of $\mathbb{N}$ there exists a least element $r \in S$. Since $r \in S$, we have $r = a - bq$ for some $q \in \mathbb{Z}$ and $r \geq 0$, so it remains to show that $r < |b|$. If we suppose by contradiction that $r \geq |b|$, then $r - |b| \geq 0$ and:

$$\begin{aligned} r - |b| &= a - bq - |b| \\ &= a - bq - \operatorname{sgn}(b) \cdot b \\ &= a - b(q + \operatorname{sgn}(b) \in S \end{aligned}$$

which contradicts r being the least element in S.

We now show that r and q are unique. Suppose by contradiction, that $bq' + r = a = bq + r$ for some $q, q' \in \mathbb{Z}$ and $0 \leq r, r' < |b|$. Suppose without loss of generality that $r \geq r'$. Then $r - r' = bq' - bq = b(q' - q)$, so $b(q' - q) \geq 0$. If $q' - q = 0$ then $r - r' = 0$ and we are done. Otherwise, if $q' - q \neq 0$, then $|q' - q| \geq 1$ and so:

$$\begin{aligned} r &= r' + b(q' - q) \\ &= r' + |b||q' - q| \\ &\geq r' + |b| \\ &\geq |b| \end{aligned}$$

which contradicts that $0 \leq r' < |b|$. ■

We often denote the remainder r of two integers a and b as $\text{rem}(a, b)$.

Example 5.2.5. Let $a = -7$ and $b = 3$. By Theorem 5.1.4, there exists unique q and $r \in \mathbb{Z}$ such that:

$$-7 = 3q + r$$

where $r \in [0, |b|)$. Indeed, the unique representation is:

$$-7 = 3 \cdot (-3) + 2$$

This implies $\text{rem}(-7, 3) = 2$.

Remark 5.2.6. $\text{rem}(a, b) = \text{rem}(a, -b)$ since $a = bq + r = (-b)(-q) + r$. However, $\text{rem}(a, b) \neq \text{rem}(-a, b)$. Indeed, $\text{rem}(7, 3) = 1$ because $7 = 3 \cdot 2 + 1$, but as seen in Example 5.1.5, $\text{rem}(-7, 3) = 2$.

Corrolary 5.2.7

Let $a, b \in \mathbb{Z}$ and $b \neq 0$. Then $b \mid a$ if and only if the remainder of a and $b = 0$.

Proof. For the forwards direction, assume $b \mid a$, then by definition we have $a = bq$ for some $q \in \mathbb{Z}$. Thus $a = bq + 0$, so by the uniqueness of such a representation by Theorem 5.1.4, the remainder is zero. Conversely, suppose the remainder is zero, then for $a = bq + r$, if $r = 0$, $a = bq$ which immediately implies $b \mid a$. ■

6 The Greatest Common Divisor

6.1 The GCD

Definition 6.1.1

Let $a, b \in \mathbb{Z}$ such that a and b are nonzero. The **greatest common divisor** (often abbreviated **GCD**) **of** a **and** b, denoted $\gcd(a, b)$ is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Theorem 6.1.2 (Bézout's Identity)

Let $a, b \in \mathbb{Z} \setminus \{0\}$ and set $d := \gcd(a, b)$. Then every common divisor of a and b also divides d. Furthermore, there exists $u, v \in \mathbb{Z}$ such that:

$$d = ua + vb$$

Moreover, d is the smallest positive integer of the form $d = ua + vb$.

Proof. We first prove $d = ua + vb$. Let $S := \{ma + nb : m, n \in \mathbb{Z}, ma + nb \geq 1\}$. Note that $S \neq \emptyset$ because $a \operatorname{sgn}(a) + b \operatorname{sgn}(b) = |a| + |b| \geq 1$. By the well ordering property of $\mathbb{N}$, there is a least element $ua + vb \in S$. In particular, define $D := ua + vb \geq 1$. Since $d \mid a$ and $d \mid b$, $d \mid D$, thus $d \leq D$. To show $D \leq d$, it suffices to show that $D \mid a$ and $D \mid b$ by the definition of the GCD. Write $a = Dq + r$ for $r \in [0, D)$, then we write:

$$\begin{aligned} r &= a - Dq \\ &= a - uqa = vqb \\ &= (1 - uq)a - vqb \end{aligned}$$

But $r < D$ so if $r \neq 0$, then $r \in S$, which contradicts the choice of D. By the same argument, $D \mid b$ and $D \mid a$, so $D \leq d$ which implies $D = d$.

Now let $d' \mid a$ and $d' \mid b$ for some $d' \in \mathbb{Z}^+$. Then $d' \mid d$ since $d = ua + vb$. ■

Corrolary 6.1.3

For $a, b, c \in \mathbb{Z}$, $a \neq 0$, if $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Proof. We have that $1 = ua + vb$ for some $u, v \in \mathbb{Z}$, then $c = uac + vbc$ so $a \mid c$ since $a \mid uac$ and $a \mid vbc$. ■

6.2 The Euclidean Algorithm

How do we find the GCD of two integers a and b. One way is to brute force through all positive integers less than or equal to $\min(|a|, |b|)$, but this would take $\min(|a|, |b|)$ steps. A much more effective algorithm, which takes $\log_2 \min(|a|, |b|)$ steps, is known as the *Euclidean Algorithm* and was discovered over 2300 years ago.

Here is how we use the Euclidean algorithm to find the GCD of two positive integers a, b with $a \geq b$. We write:

$$\begin{aligned} a &= bq_0 + r_0, & 0 < r_0 < b \\ b &= r_0 q_1 + r_1, & 0 < r_1 < r_0 \\ r_0 &= r_1 q_2 + r_2 & 0 < r_2 < r_1 \\ &\vdots \\ r_{t-1} &= r_t q_{t+1} + r_{t+1} \end{aligned}$$

Let $t \in \mathbb{N}$ such that $r_{t+1} = 0$. Then $r_t = \gcd(a, b)$

Example 6.2.4. Find $\gcd(111, 51)$ using the Euclidean algorithm.

Solution. We write:

$$\begin{aligned} 111 &= 2 \cdot 51 + 9 \\ 51 &= 5 \cdot 9 + 6 \\ 9 &= 1 \cdot 6 + 3 \\ 6 &= 2 \cdot \mathbf{3} + 0 \end{aligned}$$

Hence $\gcd(111, 51) = 3$. We can also use this algorithm to write 3 as a linear combination of 111 and 51 (in the form $3 = ua + vb$ using backwards substitution as follows:

$$\begin{aligned} 3 &= 9 - 6 \\ &= 9 - 51 + 9 \cdot 5 \\ &= 9 \cdot 6 - 51 \\ &= 6 \cdot 111 - 2 \cdot 51 \cdot 6 - 51 \\ &= 6 \cdot 111 - 11 \cdot 51 \end{aligned}$$ ✓

We can prove that the Euclidean algorithm returns $\gcd(a, b)$ by using recursion. Write:

$$\mathrm{Euc}(a, b) := \begin{cases} b \text{ if } b \mid a \\ \mathrm{Euc}(b, \mathrm{rem}(a, b)) \text{ otherwise} \end{cases}$$

where $\mathrm{rem}(a, b)$ is the remainder of a and b. It suffices to prove the following lemma:

Lemma 6.2.5

For $a, b \in \mathbb{N}^+$, with $a \geq b$, $\gcd(a, b) = \gcd(b, \mathrm{rem}(a, b))$.

Proof. Let $r := \mathrm{rem}(a, b)$, so $a = bq + r$ for some $q \in \mathbb{Z}$. Then $\gcd(b, r) \mid a$ because a is an integer combination of b and r. Since we also have $\gcd(b, r) \mid b$, we get $\gcd(b, r) \mid \gcd(a, b)$. Similarly, since $r = a - bq$, $\gcd(a, b) \mid r$. Since $\gcd(a, b) \mid b$ we finally get $\gcd(a, b) \mid \gcd(b, r)$. ■